Training dynamics of tree tensor networks: summary of results

Short version of the full report (`report/main.pdf`)

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Abstract

We study gradient flow on tree tensor networks (TTNs) trained to fit a teacher tensor that is a sum of rank-one components, with most of the effort on the saddle-to-saddle regime reached from a small initialization scale σ. The main result is that after the first component is learned, the length of each plateau is set by a graph quantity, the number k of cores in the smallest subtree that must change to add the next component: no plateau for $k = 1$, a plateau $\propto \ln(1/\sigma)$ for $k = 2$, and $\propto \sigma^{-(k-2)}$ for $k \geq 3$. We prove that k is exactly the order of the saddle under a spectral condition, and that the order can be lower when the condition fails. We also give a criterion, computed from the initialization, for which runs stall permanently, and we map the lazy and rich regimes.

1 Setting

Model. A TTN represents a tensor $W \in (\mathbb{R}^d)^{\otimes N}$ by K tensors G_v, called *cores*, placed on the nodes of a tree; θ denotes all core entries. Each of the N tensor modes (*sites*) is an open leg of one core. Each tree edge e is a *bond*, an index of dimension χ_e (all equal to χ unless stated) shared by the two cores it joins, and W is the contraction over all bonds. Cutting bond e splits the sites into two groups; the matricization of W along that split has rank at most χ_e. We use three trees: the MPS (a chain, one site per core, $K = N$), the binary tree (sites in pairs on the bottom cores, $K = N - 1$) and the hierarchical Tucker (HT) tree (one leaf core per site with a binary tree above, $K = 2N - 1$; internal cores carry no sites). For inputs $x_0, \dots, x_{N-1} \in \mathbb{R}^d$ the network computes $f(x) = \langle W,\, x_0 \otimes \cdots \otimes x_{N-1} \rangle$.

Task. Labels come from a teacher tensor W^* of the form

$$y = \langle W^*, x_0 \otimes \cdots \otimes x_{N-1} \rangle, \qquad W^* = \sum_{r=1}^{R} s_r\, U_r, \qquad U_r = u_0^{(r)} \otimes \cdots \otimes u_{N-1}^{(r)}, \tag{1}$$

with unit vectors $u_n^{(r)}$ and weights $s_1 > s_2 > \cdots > 0$. We call U_r the *components*. In a *pattern teacher* every $u_n^{(r)}$ is a standard basis vector, so U_r is labelled by a string such as 00011000. For whitened Gaussian inputs the population squared loss is, up to a constant,

$$L(\theta) = \tfrac{1}{2} \| W(\theta) - W^* \|_F^2. \tag{2}$$

L is a polynomial of degree $2K$ in the cores: a deep linear network (DLN) whose factors are arranged on a tree rather than a chain.

Training. Gradient flow $\dot\theta = -\nabla L(\theta)$ from fan-in Gaussian cores of scale σ, $G_v = \sigma Z_v / \sqrt{F_v}$ with Z_v standard Gaussian and F_v the fan-in of core v, so $\|W\| \sim \sigma^K$ at initialization. As in DLNs, gradient flow conserves on every bond e, between a core v and its parent p, the imbalance $Q_e = G_v G_v^\top - G_p G_p^\top$ (both cores matricized with the e-leg as rows).

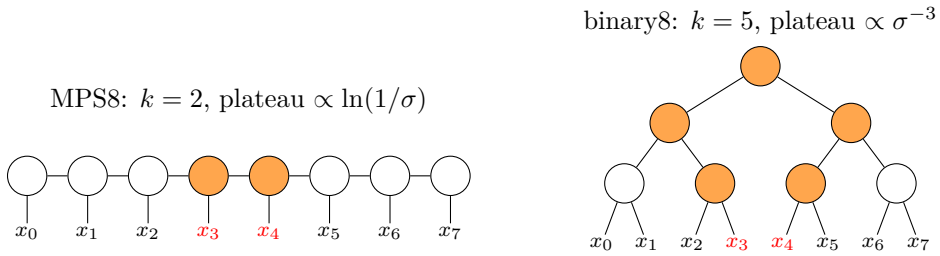


Figure 1: Routes for $j = 00000000$ learned and $r = 00011000$, so $D = \{3, 4\}$ (red). Orange: the route $\mathrm{St}(V_D)$. On HT8 the same D gives $k = 7$. Measured plateau exponents: logarithmic on MPS8, 2.998 ± 0.078 on binary8, 5.74 ± 1.06 on HT8 (limited range).

Routes. Let component j be learned and r not. Let D be the set of sites where the patterns of r and j differ, V_D the set of cores carrying them, and $\mathrm{St}(V_D)$ the Steiner subtree of V_D (the smallest subtree containing it). We call $\mathrm{St}(V_D)$ the *route* of r and write

$$k = |\mathrm{St}(V_D)|, \qquad k^* = \min_{r \text{ not learned}} k. \tag{3}$$

Figure 1 gives an example. The *order* of a saddle θ_S is the largest p with $L(\theta) - L(\theta_S) \geq -C\|\theta - \theta_S\|^p$ near θ_S; order 2 is a strict saddle.

Status labels. [proven]: theorem with proof; [derived]: calculation under stated approximations (typically leading order in σ); [empirical]: measured; [conjecture]: not established. Novelty labels: new (no prior statement found in a literature search of about 270 queries), known mechanism, new for TTNs, known. Each numerical result was re-checked by an independent re-analysis, except the test marked in Result 3.

2 Results

1. First escape

known mechanism, new for TTNs

Near the origin $L \approx \frac{1}{2}\|W^*\|^2 - g(\theta)$ with $g = \langle W, W^* \rangle$, homogeneous of degree K. Writing $\theta = \rho\hat{\theta}$ and $d\tau = \rho^{K-2}dt$, the direction $\hat{\theta}$ follows a σ-independent spherical flow and $d\ln\rho/d\tau = K\, g(\hat{\theta})$. Let t_1 be the time at which the first component is learned (its overlap with W crosses $\frac{1}{2}$). Then, for $\theta_0 = \sigma Z$,

$$t_1 = \sigma^{-(K-2)}\, \mathcal{T}(Z) + O(1), \qquad \mathcal{T}(Z) = \|Z\|^{-(K-2)} \int_0^\infty e^{-K(K-2)G(\tau)} d\tau, \quad G(\tau) = \int_0^\tau g(\hat{\theta})\, d\tau' \tag{4}$$

[proven at leading order; offset derived]. $\mathcal{T}(Z)$ is computed from the initialization by one integration before training; it predicts t_1 in 231 initialization–teacher pairs (1,848 runs, five trees) to a relative $4 \cdot 10^{-8}$–$2 \cdot 10^{-3}$ at the smallest σ, and the first component learned in 1,847 of 1,848 runs [empirical]. The norm-based time $t_{\mathrm{KH}} = K^{K/2-1}\|\theta_0\|^{-(K-2)}/((K-2)s_1)$ is a lower bound [proven]; median $\mathcal{T}/t_{\mathrm{KH}}$ ranges from 3 (binary4) to 2650 (binary8) and reaches $2.9 \cdot 10^7$ on binary16, because most of $\|\theta_0\|$ lies in core entries orthogonal to the teacher's vectors, which do not move at leading order. After the escape each plateau sits at the critical point of the problem with bond ranks fixed, and each escape raises the rank of every bond on the route by one (497 of 531 runs with recorded spectra) [empirical]. *Prior work:* the exponent $K - 2$ and the blow-up identity hold for homogeneous models (Kumar and Haupt, 2025a; Kunin et al., 2025); t_{KH} is from Kumar and Haupt (2025b). New: the evaluation on trees, the run-by-run tests and the size of the gap to t_{KH}.

2. Plateau length after the first escape: the route rule **new**

Let θ_S be the balanced critical point representing $s_j U_j$: every core is $s_j^{1/K}$ times the pattern of j on its sites, and each bond uses one channel. Entries on the other $\chi_e - 1$ channels are called *new*. For a perturbation δ of the cores and a set $\mathcal{J}$ of cores, let $W_{\mathcal{J}}[\delta]$ be W with the cores in $\mathcal{J}$ replaced by their perturbations and the others kept at θ_S; by multilinearity $W(\theta_S + \delta) = \sum_{\mathcal{J}} W_{\mathcal{J}}[\delta]$. If δ has only new entries, $W_{\mathcal{J}}[\delta] \neq 0$ requires every new index to be contracted inside $\mathcal{J}$, so a connected set $\mathcal{C}$ changes W by a product of $|\mathcal{C}|$ small entries and pays no Gauss–Newton cost at that order. Its first-order gain is $\langle \mathcal{R}, W_{\mathcal{C}}[\delta] \rangle$ with $\mathcal{R} = W^* - s_j U_j$, which depends on $\mathcal{R}$ only through its contraction with the learned cores outside $\mathcal{C}$. For a pattern teacher this contraction is

$$\mathcal{R}_{\mathcal{C}} = s_j^{(K-|\mathcal{C}|)/K} \sum_{r:\, V_{D_r} \subseteq \mathcal{C}} s_r \, U_r\big|_{\mathcal{C}}, \tag{5}$$

where D_r is the set of sites where r differs from j and $U_r|_{\mathcal{C}}$ is U_r restricted to the sites of $\mathcal{C}$ [proven], so the cheapest route to component r is $\mathrm{St}(V_{D_r})$, with coupling $c_r = s_r s_j^{(K-k)/K}$, and the saddle order is at most k^*. If the order equals k^* (Result 3), the common size b of the new entries on a balanced route obeys $\dot{b} \approx c_r b^{k-1}$. For two components learned in weight order, the time $\Delta t_2 = t_2 - t_1$ between the two learning events is

$$\Delta t_2 \simeq \begin{cases} O(1), & k = 1, \\ \dfrac{1 - s_2/s_1}{\lambda_2} \ln \dfrac{1}{\sigma}, \quad \lambda_2 = s_2 \, s_1^{(K-2)/K}, & k = 2, \\ \dfrac{1 - s_2/s_1}{(k-2)\, s_2 \, s_1^{(K-k)/K}} \, \sigma^{-(k-2)}, & k \geq 3 \end{cases} \tag{6}$$

[derived; the prefactor for $k \geq 3$ assumes an initialization aligned with the teacher's vectors]. The factor $1 - s_2/s_1$ is a head start: the route grows during the first plateau. For $k = 2$ the saddle is strict with $\lambda_{\min} = -\lambda_2$; for $k \geq 3$ the Hessian has no negative eigenvalue.

Tests [empirical]. About 11,600 runs of two-component teachers on nine trees ($K = 3$ to 15). The per-seed exponent of Δt_2 is $k - 2$ for $k = 3, 4, 5$ (named exceptions in the report), and for $k = 6, 7$ on binary8 and HT4 with balanced initialization. For $k = 2$ a log law fits better than a power law in all 14 (tree, D) settings, with slope 0.98–1.01 against the predicted 1 (head start included) and 2.00–2.07 against 2 when started at the saddle. The second bond singular value stays below $4.3\,\sigma^2 s_2$ on every bond off the route in all 5,131 learning events. Because k depends on the tree, a smaller component with a shorter route is learned first as $\sigma \to 0$: teacher 0000 (1), 0110 (0.3), 1001 (0.6) on MPS4 learns 0110 ($k = 2$) before 1001 ($k = 4$) in every run, while binary4 ($k = 3$ for both) learns them in weight order. With three or four components the rule, applied at the rank-constrained optimum reached at each plateau, predicts the plateau levels and their exponents within 0.2 (seven teachers; one exception).

Prior work: the trichotomy $O(1)$ / logarithmic / $\sigma^{-(k-2)}$ for k small layers (Rawal and DeWeese, 2026, Thm 4.2); the $k = 2$ head start in two-layer linear networks (Zhang et al., 2026, Thm 4); a descent curve along a whole connected set of rank-deficient bonds (Furman et al., 2026, Lemma D.8). New: that the relevant set is the Steiner subtree of the differing sites, which can be much smaller than the rank-deficient set; the coupling c_r; the head start for $k \geq 3$; the measurements.

3. When k^* is the saddle order **new**

Result 2 gives an upper bound. For the lower bound define, for a pattern teacher, the *generalized piggyback graph* at level ℓ: its nodes are pairs (B, π) of a core set B with $|\mathrm{St}(B)| \leq \ell$ and a pattern π

on the sites of B that differs from j on every core of B; every unlearned component r adds an edge of weight s_r between $(B_1, \pi_r|_{B_1})$ and $(B_2, \pi_r|_{B_2})$ for every split $V_{D_r} = B_1 \sqcup B_2$, where $\pi_r|_B$ is the pattern of r on the sites of B. A single-core node stands for a rotation of that core's learned vector, a multi-core node for new entries on $\mathrm{St}(B)$. Let Λ_ℓ be the largest eigenvalue of its adjacency matrix $\Gamma^{(\ell)}$.

Theorem ([proven]; Appendix A.3)**.** *Let θ_S represent one learned component j, with $\chi_e \geq 2$ on the bonds of the route. (a) If $\Lambda_{\lceil k^*/2 \rceil - 1} < s_j$, the saddle order is exactly k^*. (b) If $\Lambda_\ell > s_j$ for some $\ell < k^*/2$ (and enough spare channels), the order is at most 2ℓ. Hence the order is $\min(k^*, 2\ell^*)$, with ℓ^* the first level at which Λ_ℓ exceeds s_j.*

Consequences: the order is 2 when $k^* = 2$; for $k^* \in \{3, 4\}$ it is k^* iff $\lambda_{\max}(\Gamma^{(1)}) < s_j$ (otherwise the saddle is strict); for two-component teachers learned in weight order it is k^* on every tree. It can drop below k^* at a saddle with positive semidefinite Hessian: on MPS6 with $j = 000000\,(1)$ and 110001, 110010 (0.8 each), $k^* = 5$ but $\Lambda_2 = 0.8\sqrt{2} > 1$, and the order is 4 (measured slope 4.000, constant -0.0350 against -0.0350). The descent uses new entries on cores 0 and 1 toward the pattern 110000, which is not a teacher component, combined with rotations of the learned vectors of cores 4 and 5. Two independent checks of the proofs and 44 numerical instances agree.

Dynamics. Where the order is $p < k^*$ the escape should take $\propto \sigma^{-(p-2)}$ rather than $\sigma^{-(k^*-2)}$ [derived]. A pre-registered test (about 2,800 runs) started gradient flow at θ_S plus a perturbation of size ϵ and fitted the escape time as $\propto \epsilon^{-\gamma}$. Where ϵ lies below the size at which the lower-order descent is predicted to dominate, it measured $\gamma = 2.001 \pm 0.001$ for $p = 4$ ($k^* = 5, 6$) and $\gamma = 3.996$–3.998 for $p = 6$ ($k^* = 7$); the route law $\gamma = k^* - 2$ fits 10^3–10^5 times worse. Where the fit window extends above that size the results were mixed [empirical; **not independently verified**].

Prior work: the upper bound (Furman et al., 2026); lower bounds only for DLNs (strict versus non-strict, Achour et al., 2024, Thm 7) and for one parity saddle (Furman et al., 2026, Prop. 5.3).

4. Tree shape and strict saddles **new**

With one learned component of a pattern teacher, θ_S is strict iff $k^* = 2$ or $\lambda_{\max}(\Gamma^{(1)}) > s_j$ [proven]; the Hessian block for rotations of the learned vectors of single cores is $s_j^{(K-2)/K}(s_j I - \Gamma^{(1)})$. For non-orthogonal teachers (overlapping components, misaligned input covariance, correlated sites, random TTN teachers) the first plateau sits at a rank-one critical point $\bar{s}\, y_0 \otimes \cdots \otimes y_{N-1}$ (unit vectors y_n) that is not a teacher component. Criticality removes every single-core direction, so a two-core route has a nonzero residual only if both cores carry sites. On an MPS every bond joins two such cores, and bond $e = (i, i+1)$ has Hessian eigenvalue $-\bar{s}^{(N-2)/N} s_2(M_e)$, where M_e is the $d \times d$ matrix obtained by contracting W^* with y_n on every site $n \neq i, i+1$, and s_2 is its second singular value; the second escape takes $h_e \ln(1/\sigma)$ with $h_e = (1 - s_2(M_e)/\bar{s})/(\bar{s}^{(N-2)/N} s_2(M_e))$ [derived], which matches the measurements within 1–1.6%. On binary trees with $N = 2^n$ and on HT trees no two adjacent cores both carry sites, so $k^* \geq 3$ and the escape stays polynomial (13 teachers, $N = 4, 8$) [derived, empirical]. *Prior work:* for DLNs the piggyback graph is a single edge (Achour et al., 2024, Thm 7).

5. Order of learning **known mechanism, new for TTNs**

Let two components differ on sites carried by m cores. For $m \geq 3$ every component is a local maximum of g on the sphere; on an MPS with an orthogonal teacher the components evolve independently at leading order, so component r blows up at $\sigma^{-(K-2)} \hat{t}_r / s_r$, where $\hat{t}_r$ is the blow-up time of component r alone at unit weight and scale (as $\mathcal{T}$ in Result 1); the $\hat{t}_r$ are i.i.d., and

$$P(\text{larger first}) = P(\hat{t}_2/\hat{t}_1 > s_2/s_1), \tag{7}$$

independent of σ [proven at leading order]. All 40,800 MPS runs of our sweep learn the winner of this race first; overall P(larger first) = 0.60–0.88 at $s_2/s_1 = 0.5$ (67,200 runs) [empirical]. For $m = 2$ the smaller component is a saddle of the spherical flow with n unstable directions ($n = 1$ if the two cores are not adjacent, $n = \chi_e$ if they share bond e), and P(smaller first) $\propto \sigma^\gamma$, $\gamma = n(1-w)/w$, $w = s_2/s_1$: consistent with the data for $n = 1$ [empirical, heuristic mechanism], a conjecture for adjacent cores. *Prior work:* greedy learning in DLNs (Jacot et al., 2022, Thm 2); the tensor/matrix dichotomy and the rule that the winner maximizes s_r times the $(K-2)$-th power of its initial overlap, in multi-spike tensor PCA (Ben Arous et al., 2025). New: the classification by m, the exact race on MPS, the σ^γ law.

6. Permanent stalls: a criterion from the initialization **new**

Of 5,682 two-component runs, 398 stall permanently, all with $k \geq 4$; with χ at the teacher rank most runs with $K = 7, 8$ stall [empirical]. The stalled runs have a route-end core whose route entry has decayed to roundoff. Let u be an end of the route (it carries a differing site) and w its neighbour on the route, sharing bond e. Let $X_u^\perp$, $X_w^\perp$ be their entries on the unused channels of e with physical index orthogonal to the learned pattern. The *route entry* of u, its entry on the route's channel along the pattern of r, is one of them, since u carries a site where r differs from j. On the plateau these blocks produce a two-core term that the residual does not contain, so the loss contains a term $\frac{\kappa}{2}\|X_u^{\perp\top} X_w^\perp\|^2$ with rows of X indexed by the unused channels and $\kappa > 0$ set by the learned cores. The flow on this term is a two-layer linear network with zero target. It conserves

$$C_u = X_u^\perp X_u^{\perp\top} - X_w^\perp X_w^{\perp\top}, \tag{8}$$

and $X_u^\perp X_u^{\perp\top} \to [C_u]_+$. With one unused channel, the route entry of u survives iff $C_u > 0$ [proven in the reduced flow]. The decay rate is $\propto \sigma^2$ (σ^4 at a binary or HT leaf, whose neighbour carries no site), while the escape takes $\sigma^{-(k-2)}$; so routes with $k \leq 3$ never die, for $k = 4$ decay and escape are of the same order and a reduced model of the route decides, and for $k = 5$ a route dies when two or more of its ends die, while a single dying end leaves it stuck at an amplitude $\propto \sigma^2$ [derived]. Because the state at the start of the plateau is σ times a function of the initial direction, C_u is computed from the initialization. For a run started at θ_S plus fan-in noise, an end dies with probability $I_{F_u/(F_u+F_w)}(n_u/2, n_w/2)$ (I the regularized incomplete beta function, n the number of orthogonal physical directions): 0.392 at an MPS chain end, 0.5 at a root end ($d = \chi = 2$) [proven]. On the 3,492 fan-in runs with $k \geq 4$, 93% of runs predicted dead (by the sign of C_u at their route ends) stall, against 1.8% of those predicted alive [empirical]. Balanced initialization ($Q_e = 0$) protects ends that are tree leaves and cuts stalls tenfold. Derivation: Appendix A.4. *Prior work:* a conserved quantity decides which side of a unit can vanish (Kunin et al., 2025, Lemma B.1); imbalance can create spurious minima (Furman et al., 2026, Thm G.1).

7. Lazy and rich regimes **new**

The tangent kernel at a generic initialization has rank $\sum_v F_v\chi_v - \sum_e \chi_e^2$ (χ_v the dimension of the bond from v to its parent, 1 at the root; capped at d^N), the dimension of the fixed-rank manifold; it reaches d^N iff every bond is full, $\chi_e \geq \min(d^{n_e}, d^{N-n_e})$ with n_e the number of sites below e [proven]. Lazy training therefore reaches zero loss only with full bonds; otherwise it stops at $\frac{1}{2}\|P_\perp W^*\|^2$, with $P_\perp$ the projection onto the orthogonal complement of the kernel's range. We map regimes with the initial function subtracted ($f_\theta - f_{\theta_0}$). On binary4 with full bonds, training is saddle-to-saddle below $\sigma_c \approx 0.3\,\chi^{-1/2}$ [empirical] and lazy above $\sigma_{\text{lazy}} \approx 6.5\,\chi^{-1/2}$ [derived; measured slope -0.504 ± 0.011]; the exponent $-\frac{1}{2}$ comes from the second-order term of the kernel change, which dominates the first-order term at large χ. The saddle-to-saddle boundary rises with K (above $\sigma = 1$ for most seeds at $K \geq 8$); its large-K limit is open. We also give a μP-type parameterization that keeps the kernel moving as χ

grows [derived]. *Prior work:* regime maps for linear and diagonal networks (Woodworth et al., 2020; Kunin et al., 2024).

8. Optimizers

- known mechanism, new for TTNs GD is linearly stable at the balanced global minimum iff $\eta < \eta_c = 2/(Ks_1^{2-2/K})$, since $\lambda_{\max}(\nabla^2 L) = Ks_1^{2-2/K}$ there [proven]. Below η_c GD follows the saddle sequence of gradient flow (765 of 765 runs) with an $O(\eta)$ shift of the event times [derived, empirical]. DLN case: Mulayoff and Michaeli (2020).
- known mechanism, new for TTNs SGD does not shorten the first plateau; it lengthens later plateaus and can trap routes [empirical].
- known Adam's first escape takes a σ-independent number of steps until ϵ dominates [derived, empirical]; concurrent with Sabanayagam and Lucey (2026).
- ALS and DMRG are invariant to σ [proven]; they and Riemannian GD have no plateaus [derived].

9. MNIST new

On 4×4 MNIST images with a binary tree over the pixels ($N = 16$), the first route learned is a pair of sibling bottom cores ($k = 3$). For a pair of bottom cores, the route coupling is the norm of the label residual's correlation with that pair's features at the best rank-one fit. In a pre-registered test over 27 pixel orderings, the share of squared route couplings on pairs whose route has $k \leq 5$, computed from the data before training, predicted the ranking of the orderings by training loss at a fixed time (Spearman -0.60, 95% CI $[-0.82, -0.22]$; -0.55 within the 20 random orderings) but not by time to reach loss 0.25 [empirical]. Orderings that are equally local in the image but pair pixels vertically rank 13–21 of 27, so spatial locality alone does not explain the effect. That the ordering acts by placing label-relevant pixel pairs on routes short enough to escape within the training time is a [conjecture].

3 Already known in this setting

The binary tree with $N = 4$ is exactly a depth-3 DLN, so its saddle sequence, greedy limit and strict wrong-order saddles are known (Saxe et al., 2014; Jacot et al., 2022; Achour et al., 2024). The conservation of $Q_e = G_v G_v^\top - G_p G_p^\top$ on every bond is the tree form of DLN balancedness; the balanced, minimum-norm escape direction from the origin is in Kumar and Haupt (2025a); the two-core TTN is matrix factorization (Jiang et al., 2023); low hierarchical-rank bias of hierarchical tensor factorization is in Razin et al. (2022). We found no prior analysis of saddle-to-saddle dynamics on TTNs with $K \geq 3$ dense cores.

4 Limitations

Small trees ($N \leq 16$, mostly $N \leq 8$), exact population loss with whitened inputs, and mostly synthetic teachers. The saddle-order theorem covers saddles with one learned component. The large-K boundary of the saddle-to-saddle regime is open. The dynamics test of Result 3 was not independently re-checked.

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A Derivations

Throughout, θ_S is a balanced critical point representing $s_j U_j$ (one learned component). Each core then equals $q_j = s_j^{1/K}$ times the pattern of j on its sites and channel 1 on its bonds. $\mathcal{R} = W^* - s_j U_j$ is the residual, $L_S = \frac{1}{2}\|\mathcal{R}\|^2$, and entries on bond channels other than 1 are *new*.

A.1 Escape from the origin (Result 1)

Near the origin $\dot\theta = \nabla g(\theta) + O(\|\theta\|^{2K-1})$. Since g is homogeneous of degree K, $\nabla g(\rho\hat\theta) = \rho^{K-1}\nabla g(\hat\theta)$ and $\langle\hat\theta, \nabla g(\hat\theta)\rangle = K g(\hat\theta)$. Hence $\dot\rho = \rho^{K-1} K g(\hat\theta)$ and $\dot{\hat\theta} = \rho^{K-2}(I - \hat\theta\hat\theta^\top)\nabla g(\hat\theta)$. In the time $d\tau = \rho^{K-2}dt$ the direction obeys a ρ-free equation and $\rho(\tau) = \rho_0 e^{KG(\tau)}$, so $dt = \rho_0^{-(K-2)} e^{-K(K-2)G(\tau)} d\tau$, and ρ blows up at $t^* = \rho_0^{-(K-2)} \int_0^\infty e^{-K(K-2)G} d\tau$. With $\rho_0 = \sigma\|Z\|$ this is (4). The direction converges to a KKT point of g on the sphere, which is balanced with core norms $1/K$; the $O(1)$ offset in (4) is the time a balanced rank-one mode takes from there to the event.

A.2 Routes and the Steiner identity (Result 2)

For a set $\mathcal{J}$ of cores, let $W_{\mathcal{J}}[\delta]$ be W with the cores in $\mathcal{J}$ replaced by perturbations δ_v and the rest at θ_S. Then $W(\theta_S + \delta) = \sum_{\mathcal{J}} W_{\mathcal{J}}[\delta]$ and

$$L(\theta_S + \delta) = L_S - \sum_{\mathcal{J}\neq\emptyset} \langle \mathcal{R}, W_{\mathcal{J}}[\delta]\rangle + \tfrac{1}{2}\Big\| \sum_{\mathcal{J}\neq\emptyset} W_{\mathcal{J}}[\delta]\Big\|^2. \tag{9}$$

A new index of δ_v contracts, in a neighbouring unperturbed core, with a zero block. So $W_{\mathcal{J}}[\delta] \neq 0$ with only new entries requires every new index to be matched inside $\mathcal{J}$: each connected part $\mathcal{C}$ of $\mathcal{J}$ has at least two cores, and $W_{\mathcal{C}}[\delta] = \mathcal{W}_{\mathcal{C}}(\delta) \otimes \bigotimes_{f\in\partial\mathcal{C}} \Sigma_f$, where $\mathcal{W}_{\mathcal{C}}$ is the small TTN formed by the δ_v and Σ_f is the learned network beyond the boundary bond f. For a proper subset of $\mathcal{C}$ the product vanishes, so the gain $\langle \mathcal{R}, W_{\mathcal{C}}\rangle$ is of order $|\mathcal{C}|$ in δ while the quadratic term is of order $2|\mathcal{C}|$.

Beyond f the learned network is $\Sigma_f = q_j^{n_f} U_j|_{H_f}$ (n_f cores, sites H_f). Contracting $\mathcal{R} = \sum_{r\neq j} s_r U_r$ with it gives zero unless U_r agrees with U_j on H_f, i.e. unless all sites where r differs from j lie on cores of $\mathcal{C}$. This gives (5), and a connected $\mathcal{C} \supseteq V_{D_r}$ contains $\mathrm{St}(V_{D_r})$. Along the curve where each route core gets t times the pattern of r on a new channel, $L - L_S = -c_r t^k + \frac{1}{2} q_j^{2(K-k)} t^{2k}$.

Escape time. With old entries frozen and route amplitudes b_v, $\dot b_v = A \prod_{u\neq v} b_u\,(s_r - A\prod_u b_u)$, $A = q_j^{K-k}$. For a balanced route at small amplitude $\dot b = c_r b^{k-1}$, so $b^{-(k-2)}$ decreases at rate $(k-2)c_r$ (or $\ln b$ grows at rate c_r for $k=2$). During the first plateau the route already grows with coupling $s_r q_j(t)^{K-k}$ while q_j rises from σ; integrating with $\dot q_j \simeq s_j q_j^{K-1}$ gives $b_{\mathrm{eff}}^{-(k-2)} = \sigma^{-(k-2)}(1 - s_r/s_j) + O(1)$ for $k \geq 3$ and $\ln(1/b_{\mathrm{eff}}) = (1 - s_r/s_j)\ln(1/\sigma) + O(1)$ for $k = 2$, which gives (6).

A.3 Saddle order (Result 3)

Let P_v project the physical space of core v onto the pattern of j, $P_v^\perp = I - P_v$, and $\Pi_B = \bigotimes_{v\in B} P_v^\perp \otimes \bigotimes_{v\notin B} P_v$. The 2^K sector projectors are orthogonal and sum to the identity. With $\deg B = |\mathrm{St}(B)|$, $k^* = \min\{\deg B : \Pi_B \mathcal{R} \neq 0\}$. The general operator $\Gamma^{(\ell)}$ acts on $\bigoplus_{\deg B \leq \ell} \mathrm{range}(\Pi_B)$ with block $(x, y) \mapsto \langle \Pi_{B_1 \sqcup B_2}\mathcal{R}, x \otimes y\rangle$ for disjoint B_1, B_2 and zero otherwise; for a pattern teacher it is the adjacency matrix of the graph in Result 3.

Theorem (Lower bound)**.** *If $\Lambda_{\ell-1} < s_j$ for some $2 \leq \ell \leq \lceil k^*/2\rceil$, then $L(\theta) - L_S \geq -C\|\theta - \theta_S\|^{\min(k^*, 2\ell)}$ near θ_S.*

Proof sketch. Let $y = \langle U_j, W\rangle$ and $\mu_B = \Pi_B W/y$. Sector orthogonality gives exactly

$$L - L_S = \tfrac{1}{2}(y - s_j)^2 + \sum_{B\neq\emptyset}\Big[\tfrac{y^2}{2}\|\mu_B\|^2 - y\langle \Pi_B\mathcal{R}, \mu_B\rangle\Big].$$

Families (f_B) form a commutative algebra under $(f * g)_B = \sum_{B_1\sqcup B_2=B} f_{B_1}\otimes g_{B_2}$; let $\psi = \log\mu$ (set-partition cumulants). If the new entries ν_v of a core vanish, the contraction factorizes at v and no cumulant contains a set through v; analyticity turns this into $\|\psi_B\| \le C\prod_{v\in\mathrm{St}(B)}\|\nu_v\|$. Split sets into low (deg $\le \ell-1$) and high. Low sets carry no residual, the Gauss–Newton term is at least $(1 - C\varepsilon)\|\psi_{\mathrm{low}}\|^2$, and the gain terms with two low cumulants sum to $\frac{1}{2}\langle\psi_{\mathrm{low}}, \Gamma^{(\ell-1)}\psi_{\mathrm{low}}\rangle$. Every other term is $O(\varepsilon^{k^*})$ or $O(\varepsilon^{2\ell})$. Hence $L - L_S \ge \frac{y}{2}(s_j - \Lambda_{\ell-1} - O(\varepsilon))\|\psi_{\mathrm{low}}\|^2 - O(\varepsilon^{\min(k^*,2\ell)})$. □

Theorem (Sharpness)**.** *For a pattern teacher, if a node set X of the level-ℓ graph ($\ell < k^*/2$) has adjacency spectral radius $r(A_X) > s_j$ and each bond has enough spare channels, there is a curve with $\|\theta(t) - \theta_S\| \le t$ and $L - L_S \le -Ct^{2\ell}$.*

Proof sketch. Take the Perron vector x of A_X. Rotate each single-core node by εx toward its pattern (old entries) and give each multi-core node (B,π) new entries of size $q_j(\varepsilon x_{B,\pi})^{1/\deg B}$ on $\mathrm{St}(B)$, each on its own spare channel. Then $L - L_S = -\frac{s_j}{2}(r(A_X) - s_j)\varepsilon^2 + O(\varepsilon^3)$ while the distance is $O(\varepsilon^{1/\ell})$. □

For a two-component teacher in weight order every node has at most one edge, so $\Lambda_\ell \in \{0, s_r\}$ with $s_r < s_j$ and the order is k^*. In the MPS6 example the node $(\{0,1\},11)$ is joined to $(\{5\},1)$ by 110001 and to $(\{4\},1)$ by 110010: a star with $\Lambda_2 = 0.8\sqrt{2}$.

A.4 Route-end trap (Result 6)

Let u be a route end with route neighbour w on bond e, and suppose one unused channel of e is in play. Write $a = \|X_u^\perp\|^2$, $b = \|X_w^\perp\|^2$. On the plateau the old entries are at θ_S and the new entries are small, so the only term in the loss at second order in these two blocks is the Gauss–Newton cost $\frac{\kappa}{2}\|X_u^{\perp\top}X_w^\perp\|^2$, with $\kappa = q_j^{2(K-2)}$ when w carries sites: their product changes W by a tensor that differs from U_j exactly on the sites of u and w, and the residual has no such component unless some component has the two-core route $\{u, w\}$. Gradient flow on this term gives

$$\dot a = -2\kappa ab, \qquad \dot b = -2\kappa ab,$$

so $C_u = a - b$ is conserved and $\dot a = -2\kappa a(a - C_u)$. If $C_u > 0$, $a \to C_u$; if $C_u < 0$, $a \to 0$ at rate $2\kappa|C_u|$. The route entry of u lies in $X_u^\perp$ (it carries a site where r differs from j), so it survives iff $C_u > 0$. With several channels the same computation gives $X_u^\perp X_u^{\perp\top} \to [C_u]_+$.

Time scales. $|C_u| = O(\sigma^2)$, so a dying end decays in time $\propto \sigma^{-2}$, against the escape time $\sigma^{-(k-2)}$: the end is decided before the escape when $k \ge 5$, at the same order when $k = 4$, and the route escapes first when $k \le 3$. If w carries no site (binary or HT leaf), the cheapest costly product passes through three cores, the rate is $\propto \sigma^4$, and the threshold becomes $k \ge 6$.

From the initialization. The full imbalance Q_e (Section 1) is exactly conserved. Let ν be the route's channel on e and $Q_{\nu\nu} = \nu^\top Q_e \nu$, with Q_e oriented as end minus neighbour. $Q_{\nu\nu}$ is the squared norm of all of u's entries on ν minus that of w's. Splitting these entries into the orthogonal blocks and the entries that pair with nothing at second order (X_{nv} for u, Y_{nv} for w) gives $C_u = Q_{\nu\nu} + \|Y_{\mathrm{nv}}\|^2 - \|X_{\mathrm{nv}}\|^2 + O(\sigma^3)$, evaluated after the first escape. That state is σ times a function of the initial direction, obtained from the σ-free flow of A.1.

Death probability. At a saddle start the blocks are Gaussian with variances σ^2/F_u and σ^2/F_w, so $a \sim \sigma^2\chi^2_{n_u}/F_u$ and $b \sim \sigma^2\chi^2_{n_w}/F_w$. With $Z = \chi^2_{n_u}/(\chi^2_{n_u} + \chi^2_{n_w}) \sim \mathrm{Beta}(n_u/2, n_w/2)$, $a < b$ iff $Z < F_u/(F_u + F_w)$, so $P(\text{end dies}) = I_{F_u/(F_u+F_w)}(n_u/2, n_w/2)$.

B Figures (from the full report)

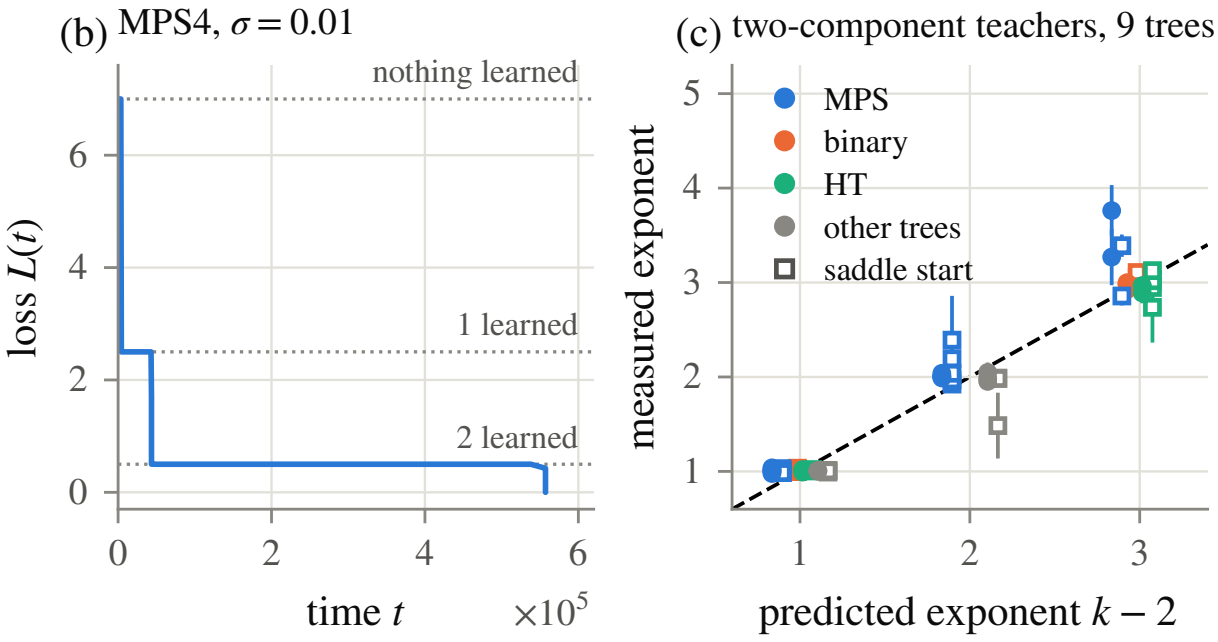


Figure 2: Saddle-to-saddle runs with the predicted plateau levels, and the measured plateau exponent against $k-2$ (full report, Fig. 1).

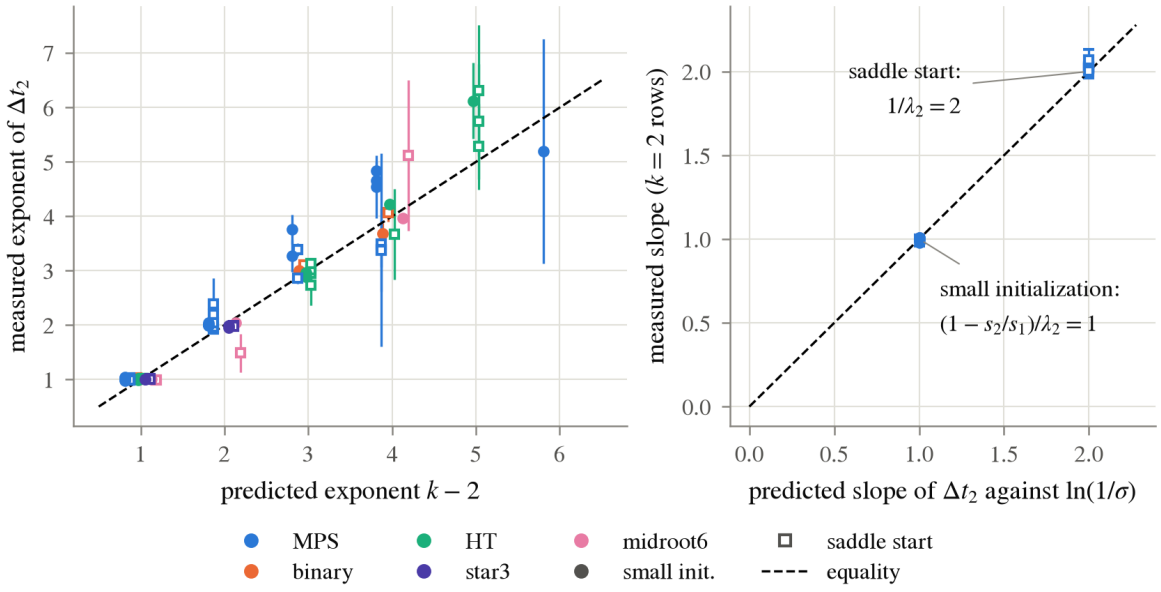


Figure 3: Per-seed plateau exponent against $k-2$ for every two-component setting with $k \geq 3$ (left), and the $k=2$ log slope against $(1-s_2/s_1)/\lambda_2$ and $1/\lambda_2$ (right) (full report, Sec. 4.4).

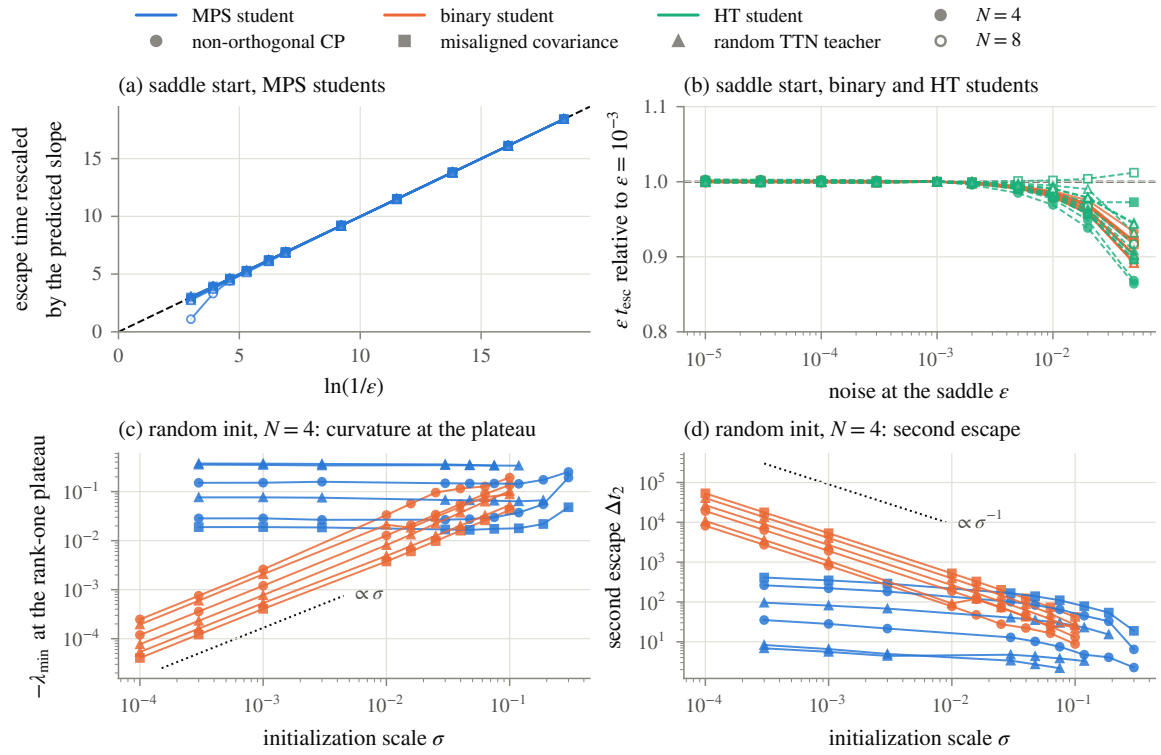


Figure 4: Non-orthogonal teachers: logarithmic escape from the rank-one saddle on an MPS, power law on binary and HT trees (full report, Sec. 6.2).

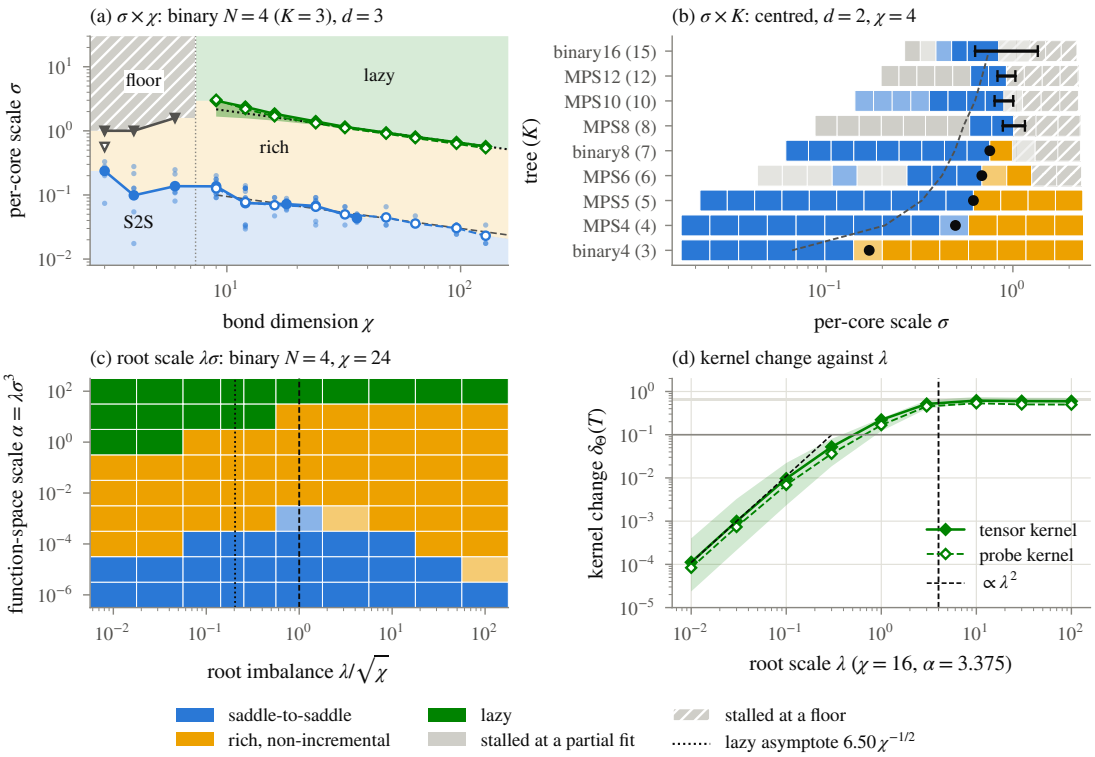


Figure 5: Regimes of binary4 in (σ, χ) (full report, Sec. 3).